# Logistic Regression and Factor Analysis on Airplane Dataset

## Motive of this project

To find which service factors drive satisfaction, then predict satisfied vs. neutral/dissatisfied with logistic regression

## Code and Interpretation

**#Required libraries**
library(readr)
library(ggplot2)
library(corrplot)
library(factoextra)
library(FactoMineR)
library(psych)
library(DescTools)
library(pROC)

**#importing the dataset and cleaning it:**
data<-read.csv("C:/Users/USER/Desktop/msc_sanii/airplane.csv")

head(data)

```
> head(data)
  X     id Gender     Customer.Type Age  Type.of.Travel    Class Flight.Distance Inflight.wifi.service
1 0  70172   Male    Loyal Customer  13 Personal Travel Eco Plus             460                     3
2 1   5047   Male disloyal Customer  25 Business travel Business            235                     3
3 2 110028 Female    Loyal Customer  26 Business travel Business           1142                     2
4 3  24026 Female    Loyal Customer  25 Business travel Business            562                     2
5 4 119299   Male    Loyal Customer  61 Business travel Business            214                     3
6 5 111157 Female    Loyal Customer  26 Personal Travel      Eco           1180                     3
  Departure.Arrival.time.convenient Ease.of.Online.booking Gate.location Food.and.drink Online.boarding Seat.comfort
1                                 4                      3             1             5               3            5
2                                 2                      3             3             1               3            1
3                                 2                      2             2             5               5            5
4                                 5                      5             5             2               2            2
5                                 3                      3             3             4               5            5
6                                 4                      2             1             1               2            1
  Inflight.entertainment On.board.service Leg.room.service Baggage.handling Checkin.service Inflight.service Cleanliness
1                      5                4                3                4               4                5           5
2                      1                1                5                3               1                4           1
3                      5                4                3                4               4                4           5
4                      2                2                5                3               1                4           2
5                      3                3                4                4               3                3           3
6                      1                3                4                4               4                4           1
  Departure.Delay.in.Minutes Arrival.Delay.in.Minutes           satisfaction
1                         25                       18 neutral or dissatisfied
2                          1                        6 neutral or dissatisfied
3                          0                        0              satisfied
4                         11                        9 neutral or dissatisfied
5                          0                        0              satisfied
6                          0                        0 neutral or dissatisfied
```

str(data)

```
> str(data)
'data.frame':   103904 obs. of  25 variables:
 $ X                                : int  0 1 2 3 4 5 6 7 8 9 ...
 $ id                               : int  70172 5047 110028 24026 119299 111157 82113 96462 79485 65725 ...
 $ Gender                           : chr  "Male" "Male" "Female" "Female" ...
 $ Customer.Type                    : chr  "Loyal Customer" "disloyal Customer" "Loyal Customer" "Loyal Customer" ...
 $ Age                              : int  13 25 26 25 61 26 47 52 41 20 ...
 $ Type.of.Travel                   : chr  "Personal Travel" "Business travel" "Business travel" "Business travel" ...
 $ Class                            : chr  "Eco Plus" "Business" "Business" "Business" ...
 $ Flight.Distance                  : int  460 235 1142 562 214 1180 1276 2035 853 1061 ...
 $ Inflight.wifi.service            : int  3 3 2 2 3 3 2 4 1 3 ...
 $ Departure.Arrival.time.convenient: int  4 2 2 5 3 4 4 3 2 3 ...
 $ Ease.of.Online.booking           : int  3 3 2 5 3 2 2 4 2 3 ...
 $ Gate.location                    : int  1 3 2 5 3 1 3 4 2 4 ...
 $ Food.and.drink                   : int  5 1 5 2 4 1 2 5 4 2 ...
 $ Online.boarding                  : int  3 3 5 2 5 2 2 5 3 3 ...
 $ Seat.comfort                     : int  5 1 5 2 5 1 2 5 3 3 ...
 $ Inflight.entertainment           : int  5 1 5 2 3 1 2 5 1 2 ...
 $ On.board.service                 : int  4 1 4 2 3 3 3 5 1 2 ...
 $ Leg.room.service                 : int  3 5 3 5 4 4 3 5 2 3 ...
 $ Baggage.handling                 : int  4 3 4 3 4 4 4 5 1 4 ...
 $ Checkin.service                  : int  4 1 4 1 3 4 3 4 4 4 ...
 $ Inflight.service                 : int  5 4 4 4 3 4 5 5 1 3 ...
 $ Cleanliness                      : int  5 1 5 2 3 1 2 4 2 2 ...
 $ Departure.Delay.in.Minutes       : int  25 1 0 11 0 0 9 4 0 0 ...
 $ Arrival.Delay.in.Minutes         : num  18 6 0 9 0 0 23 0 0 0 ...
 $ satisfaction                     : chr  "neutral or dissatisfied" "neutral or dissatisfied" "satisfied" "neutral or dissati
sfied" ...
```

## Interpretation: Data Import

- **Raw data:** 103,904 passengers and 25 variables. These are passenger details (gender, age, customer type, travel type, class, flight distance), 14 service ratings (1-5), two delay variables, and the target `satisfaction`.
- **Columns dropped later:** `X` and `id` are only an index and an identifier.

**#Factoring the characteristic columns**

```
data$Gender <- as.factor(data$Gender)
data$Customer.Type <- as.factor(data$Customer.Type)
data$Type.of.Travel <- as.factor(data$Type.of.Travel)
data$Class <-as.factor(data$Class)
data$satisfaction <- as.factor(data$satisfaction)
```

str(data)

```
> str(data)
'data.frame':   103904 obs. of  25 variables:
 $ X                                : int  0 1 2 3 4 5 6 7 8 9 ...
 $ id                               : int  70172 5047 110028 24026 119299 111157 82113 96462 79485 65725 ...
 $ Gender                           : Factor w/ 2 levels "Female","Male": 2 2 1 1 2 1 2 1 1 2 ...
 $ Customer.Type                    : Factor w/ 2 levels "disloyal Customer",..: 2 1 2 2 2 2 2 2 2 1 ...
 $ Age                              : int  13 25 26 25 61 26 47 52 41 20 ...
 $ Type.of.Travel                   : Factor w/ 2 levels "Business travel",..: 2 1 1 1 1 2 2 1 1 1 ...
 $ Class                            : Factor w/ 3 levels "Business","Eco",..: 3 1 1 1 1 2 2 1 1 2 ...
 $ Flight.Distance                  : int  460 235 1142 562 214 1180 1276 2035 853 1061 ...
 $ Inflight.wifi.service            : int  3 3 2 2 3 3 2 4 1 3 ...
 $ Departure.Arrival.time.convenient: int  4 2 2 5 3 4 4 3 2 3 ...
 $ Ease.of.Online.booking           : int  3 3 2 5 3 2 2 4 2 3 ...
 $ Gate.location                    : int  1 3 2 5 3 1 3 4 2 4 ...
 $ Food.and.drink                   : int  5 1 5 2 4 1 2 5 4 2 ...
 $ Online.boarding                  : int  3 3 5 2 5 2 2 5 3 3 ...
 $ Seat.comfort                     : int  5 1 5 2 5 1 2 5 3 3 ...
 $ Inflight.entertainment           : int  5 1 5 2 3 1 2 5 1 2 ...
 $ On.board.service                 : int  4 1 4 2 3 3 3 5 1 2 ...
 $ Leg.room.service                 : int  3 5 3 5 4 4 3 5 2 3 ...
 $ Baggage.handling                 : int  4 3 4 3 4 4 4 5 1 4 ...
 $ Checkin.service                  : int  4 1 4 1 3 4 3 4 4 4 ...
 $ Inflight.service                 : int  5 4 4 4 3 4 5 5 1 3 ...
 $ Cleanliness                      : int  5 1 5 2 3 1 2 4 2 2 ...
 $ Departure.Delay.in.Minutes       : int  25 1 0 11 0 0 9 4 0 0 ...
 $ Arrival.Delay.in.Minutes         : num  18 6 0 9 0 0 23 0 0 0 ...
 $ satisfaction                     : Factor w/ 2 levels "neutral or dissatisfied",..: 1 1 2 1 2 1 1 2 1 1 ...
```

## Interpretation: Missing values

- Gender, Customer.Type, Type.of.Travel, Class and satisfaction are converted from `chr` to factors, which `glm` needs in order to build dummy variables.
- R takes the **alphabetically first level as the reference** category:
    - Gender: Female
    - Customer.Type: disloyal
    - Type.of.Travel: Business travel
    - Class: Business
    - satisfaction: “neutral or dissatisfied” is the baseline, so the model predicts P(satisfied).

**#Checking for null values and omitting the null values**
colSums(is.na(data))

```
> colSums(is.na(data))
                              X                              id                          Gender
                              0                               0                               0
                  Customer.Type                             Age                Type.of.Travel
                              0                               0                               0
                          Class                  Flight.Distance            Inflight.wifi.service
                              0                               0                               0
Departure.Arrival.time.convenient          Ease.of.Online.booking                   Gate.location
                              0                               0                               0
                  Food.and.drink                 Online.boarding                    Seat.comfort
                              0                               0                               0
          Inflight.entertainment                On.board.service                Leg.room.service
                              0                               0                               0
                Baggage.handling                Checkin.service                Inflight.service
                              0                               0                               0
                     Cleanliness       Departure.Delay.in.Minutes         Arrival.Delay.in.Minutes
                              0                               0                             310
                    satisfaction
                              0
```

data<-na.omit(data)

## Interpretation: Missing Values

- Only `Arrival.Delay.in.Minutes` has NAs, 310 of them (0.3%).
- `na.omit` leaves **103,594 rows**. Dropping so few rows is harmless and unlikely to bias the results.

**#Removing unneccesary columns from the dataset:**
new_data <- subset(data, select = -c(X, id, satisfaction))
colnames(new_data)
dim(new_data)

```
> colnames(new_data)
 [1] "Gender"                          "Customer.Type"                   "Age"
 [4] "Type.of.Travel"                  "Class"                           "Flight.Distance"
 [7] "Inflight.wifi.service"           "Departure.Arrival.time.convenient" "Ease.of.Online.booking"
[10] "Gate.location"                   "Food.and.drink"                  "Online.boarding"
[13] "Seat.comfort"                    "Inflight.entertainment"          "On.board.service"
[16] "Leg.room.service"                "Baggage.handling"                "Checkin.service"
[19] "Inflight.service"                "Cleanliness"                     "Departure.Delay.in.Minutes"
[22] "Arrival.Delay.in.Minutes"
> dim(new_data)
[1] 103594     22
>
```

**#converting into numeric**

numeric_data <- new_data[, sapply(new_data, is.numeric)]

str(numeric_data)

```
> str(numeric_data)
'data.frame':   103594 obs. of  18 variables:
 $ Age                              : int  13 25 26 25 61 26 47 52 41 20 ...
 $ Flight.Distance                  : int  460 235 1142 562 214 1180 1276 2035 853 1061 ...
 $ Inflight.wifi.service            : int  3 3 2 2 3 3 2 4 1 3 ...
 $ Departure.Arrival.time.convenient: int  4 2 2 5 3 4 4 3 2 3 ...
 $ Ease.of.Online.booking           : int  3 3 2 5 3 2 2 4 2 3 ...
 $ Gate.location                    : int  1 3 2 5 3 1 3 4 2 4 ...
 $ Food.and.drink                   : int  5 1 5 2 4 1 2 5 4 2 ...
 $ Online.boarding                  : int  3 3 5 2 5 2 2 5 3 3 ...
 $ Seat.comfort                     : int  5 1 5 2 5 1 2 5 3 3 ...
 $ Inflight.entertainment           : int  5 1 5 2 3 1 2 5 1 2 ...
 $ On.board.service                 : int  4 1 4 2 3 3 3 5 1 2 ...
 $ Leg.room.service                 : int  3 5 3 5 4 4 3 5 2 3 ...
 $ Baggage.handling                 : int  4 3 4 3 4 4 4 5 1 4 ...
 $ Checkin.service                  : int  4 1 4 1 3 4 3 4 4 4 ...
 $ Inflight.service                 : int  5 4 4 4 3 4 5 5 1 3 ...
 $ Cleanliness                      : int  5 1 5 2 3 1 2 4 2 2 ...
 $ Departure.Delay.in.Minutes       : int  25 1 0 11 0 0 9 4 0 0 ...
 $ Arrival.Delay.in.Minutes         : num  18 6 0 9 0 0 23 0 0 0 ...
```

**#Normalizing (Scaling) the data**

scaled_data<-scale(numeric_data)

head(scaled_data)

```
> head(scaled_data)
         Age Flight.Distance Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking
1 -1.7455335    -0.731301739             0.2035200                         0.61624642              0.1737152
2 -0.9515217    -0.956911509             0.2035200                        -0.69502875              0.1737152
3 -0.8853540    -0.047453458            -0.5495679                        -0.69502875             -0.5411150
4 -0.9515217    -0.629025310            -0.5495679                         1.27188400              1.6033756
5  1.4305138    -0.977968421             0.2035200                        -0.03939116              0.1737152
6 -0.8853540    -0.009350474             0.2035200                         0.61624642             -0.5411150
  Gate.location Food.and.drink Online.boarding Seat.comfort Inflight.entertainment On.board.service
1   -1.54730443      1.3523941      -0.1856314     1.182986               1.231524        0.4792350
2    0.01798067     -1.6564793      -0.1856314    -1.849854              -1.769158       -1.8494433
3   -0.76466188      1.3523941       1.2964729     1.182986               1.231524        0.4792350
4    1.58326577     -0.9042610      -0.9266836    -1.091644              -1.018987       -1.0732172
5    0.01798067      0.6001758       1.2964729     1.182986              -0.268817       -0.2969911
6   -1.54730443     -1.6564793      -0.9266836    -1.849854              -1.769158       -0.2969911
  Leg.room.service Baggage.handling Checkin.service Inflight.service Cleanliness Departure.Delay.in.Minutes
1       -0.2671418        0.3118517       0.5497705        1.1562057   1.3059070                0.26896481
2        1.2532982       -0.5348516      -1.8210288        0.3055785  -1.7424231               -0.36067985
3       -0.2671418        0.3118517       0.5497705        0.3055785   1.3059070               -0.38691505
4        1.2532982       -0.5348516      -1.8210288        0.3055785  -0.9803406               -0.09832791
5        0.4930782        0.3118517      -0.2404960       -0.5450488  -0.2182581               -0.38691505
6        0.4930782        0.3118517       0.5497705        0.3055785  -1.7424231               -0.38691505
  Arrival.Delay.in.Minutes
1               0.07290485
2              -0.23718323
3              -0.39222727
4              -0.15966121
5              -0.39222727
6              -0.39222727
```

## Interpretation: Numeric subset and scaling

- 18 numeric variables are kept (age, distance, 14 ratings, 2 delays) and standardized to mean 0 and SD 1.
- This puts delays (minutes) and ratings (1-5) on one scale, so no variable dominates the factor analysis because of its units.
- Ordinal ratings are treated as continuous. That is a common but real assumption.

# Factor Analysis

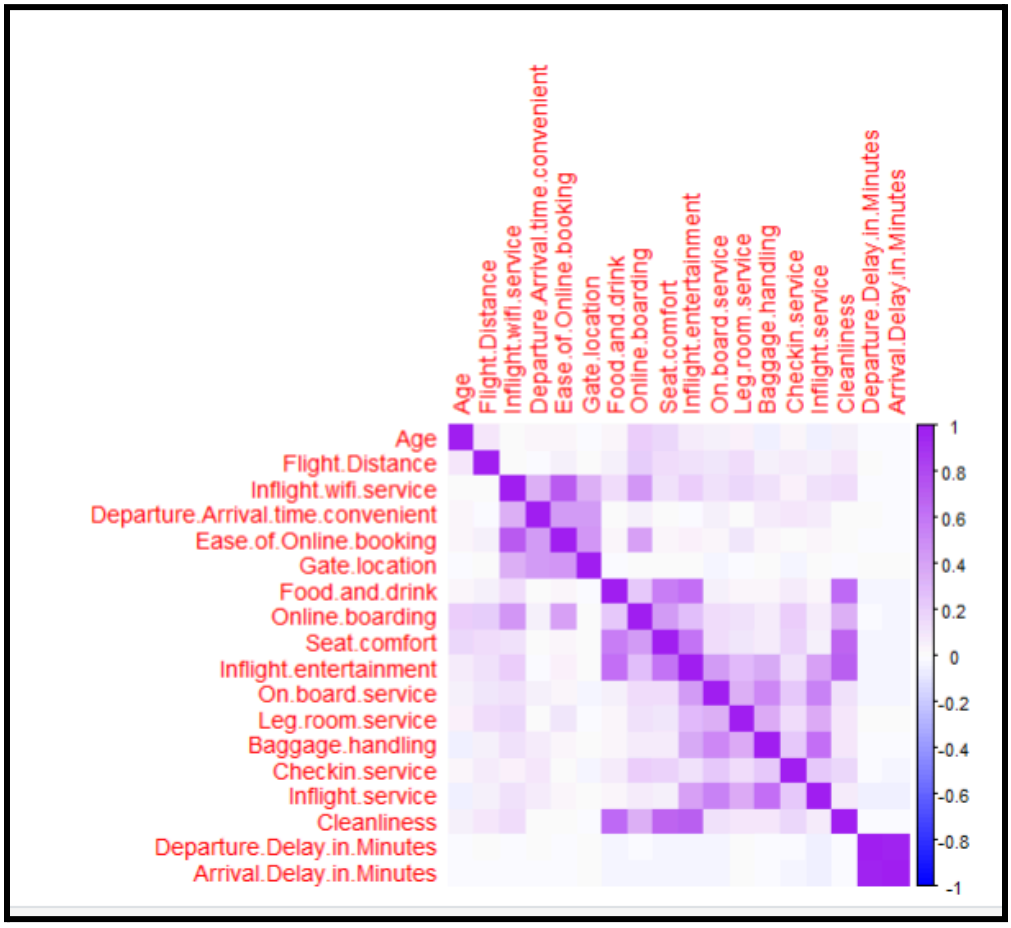
**#Covariance matrix**

```
cor_matrix<-cor(scaled_data)
cor_matrix[1:5,1:5]
```

```
> cor_matrix[1:5,1:5]
                                          Age Flight.Distance Inflight.wifi.service
Age                                1.00000000     0.099837593           0.017469851
Flight.Distance                    0.09983759     1.000000000           0.007050175
Inflight.wifi.service              0.01746985     0.007050175           1.000000000
Departure.Arrival.time.convenient  0.03803760    -0.019908273           0.343758147
Ease.of.Online.booking             0.02446145     0.065696835           0.715848109
                                   Departure.Arrival.time.convenient Ease.of.Online.booking
Age                                                       0.03803760             0.02446145
Flight.Distance                                          -0.01990827             0.06569684
Inflight.wifi.service                                     0.34375815             0.71584811
Departure.Arrival.time.convenient                         1.00000000             0.43702106
Ease.of.Online.booking                                    0.43702106             1.00000000
>
```

**#Correlation Plot**

```
corrplot(cor_matrix,
        method = "color",
        col = colorRampPalette(c("blue", "white", "purple"))(200))
```



## Interpretation: Correlation matrix and heatmap

- **Strong:** wifi and ease of online booking have r = 0.716.
- **Moderate:** ease of booking and departure/arrival convenience have r = 0.437. Wifi and departure/arrival convenience have r = 0.344.
- **Near zero:** Age and Flight Distance (r ≈ 0.10) are essentially unrelated to the other variables.
- **Visible blocks in the heatmap:**
  - online/convenience variables
  - comfort variables (food, seat, entertainment, cleanliness)
  - service variables (on-board, leg room, baggage, check-in, inflight service)
  - the two delay variables, which form a very dark pair (r ≈ 1)
- **Conclusion:** the variables clump into groups, so factor analysis is justified.

**#Kaiser-Meyer-olkin**

KMO(cor_matrix)

```
> KMO(cor_matrix)
Kaiser-Meyer-Olkin factor adequacy
Call: KMO(r = cor_matrix)
Overall MSA =  0.73
MSA for each item =
                              Age                    Flight.Distance
                             0.56                               0.70
            Inflight.wifi.service Departure.Arrival.time.convenient
                             0.73                               0.74
           Ease.of.Online.booking                      Gate.location
                             0.68                               0.71
                   Food.and.drink                    Online.boarding
                             0.84                               0.73
                     Seat.comfort          Inflight.entertainment
                             0.84                               0.77
                On.board.service                Leg.room.service
                             0.83                               0.88
                 Baggage.handling                  Checkin.service
                             0.81                               0.71
                 Inflight.service                      Cleanliness
                             0.78                               0.81
       Departure.Delay.in.Minutes          Arrival.Delay.in.Minutes
                             0.50                               0.50
>
```

**#Bartlett's test**

cortest.bartlett(cor_matrix,n=nrow(scaled_data))

```
> cortest.bartlett(cor_matrix,n=nrow(scaled_data))
$chisq
[1] 898373.3

$p.value
[1] 0

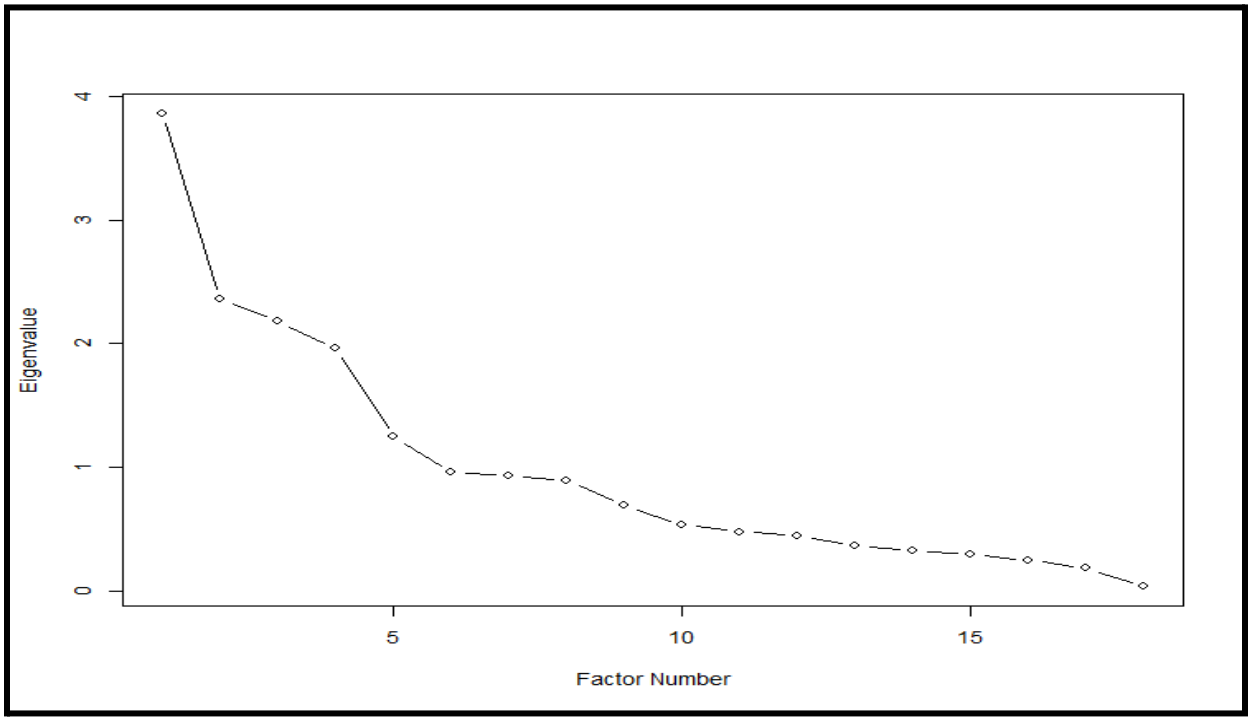
$df
[1] 153
```

## Interpretation: KMO and Bartlett's test

- **KMO overall = 0.73**, which is acceptable and above the 0.6 threshold.
- **Per-variable KMO:**
  - Best: Leg room (0.88), Food and Seat comfort (0.84).
  - Weak: Age (0.56), and both delay variables (0.50), the bare minimum.
  - The delays score low because they correlate almost only with each other.
- **Bartlett:** $\chi^2$ = 898,373, df = 153 (= 18·17/2), p ≈ 0.
  - This rejects "the correlation matrix is an identity matrix", so there is shared variance to extract.
  - With n > 100,000, Bartlett is almost always significant, so rely more on KMO.

**#Scree plot**

```
eig=eigen(cor_matrix)
plot(eig$values, type = "b",
    xlab = "Factor Number",
    ylab = "Eigenvalue")
```




## Interpretation:Scree plot

- Eigenvalues are roughly 3.85, 2.35, 2.2, 1.95, 1.25, then about 0.95 and below.
- **Kaiser rule (eigenvalue > 1):** 5 factors.
- **Elbow:** a clear drop after factor 4, then after factor 5.
- The analyst chose 4 factors for the model, but see the issues in section 12.

**#Factor analysis model**

```
fa_model=factanal(scaled_data,4,rotation = "varimax")
fa_model
```

**#view**

```
print(fa_model, digits = 3, cutoff = 0.4)
```

```
Call:
factanal(x = scaled_data, factors = 4, rotation = "varimax")

Uniquenesses:
                            Age                   Flight.Distance
                          0.986                             0.973
           Inflight.wifi.service Departure.Arrival.time.convenient
                          0.385                             0.777
          Ease.of.Online.booking                     Gate.location
                          0.115                             0.765
                 Food.and.drink                   Online.boarding
                          0.409                             0.676
                    Seat.comfort           Inflight.entertainment
                          0.390                             0.199
                On.board.service                 Leg.room.service
                          0.506                             0.753
                Baggage.handling                  Checkin.service
                          0.413                             0.916
                Inflight.service                      Cleanliness
                          0.345                             0.258
      Departure.Delay.in.Minutes          Arrival.Delay.in.Minutes
                          0.063                             0.005
```

```
Loadings:
                                Factor1 Factor2 Factor3 Factor4
Age
Flight.Distance
Inflight.wifi.service                                   0.754
Departure.Arrival.time.convenient                       0.468
Ease.of.Online.booking                                  0.939
Gate.location                                           0.484
Food.and.drink                  0.768
Online.boarding                                         0.406
Seat.comfort                    0.775
Inflight.entertainment          0.742   0.499
On.board.service                        0.698
Leg.room.service                        0.483
Baggage.handling                        0.765
Checkin.service
Inflight.service                        0.808
Cleanliness                     0.854
Departure.Delay.in.Minutes                                      0.967
Arrival.Delay.in.Minutes                                        0.997

               Factor1 Factor2 Factor3 Factor4
SS loadings      2.701   2.346   2.085   1.933
Proportion Var   0.150   0.130   0.116   0.107
Cumulative Var   0.150   0.280   0.396   0.504

Test of the hypothesis that 4 factors are sufficient.
The chi square statistic is 64966.82 on 87 degrees of freedom.
The p-value is 0
```

## Interpretation: Factor analysis with 4 factors (varimax)

| Factor | Variables (loading) | Suggested label |
|---|---|---|
| F1 | Cleanliness .854, Seat comfort .775, Food & drink .768, Inflight entertainment .742 | **Comfort and amenities** |
| F2 | Inflight service .808, Baggage .765, On-board service .698, Leg room .483, Entertainment .499 (cross-loading) | **Service and staff** |
| F3 | Ease of online booking .939, Wifi .754, Gate location .484, Dep/Arr convenience .468, Online boarding .406 | **Digital and convenience** |
| F4 | Arrival delay .997, Departure delay .967 | **Delays** |

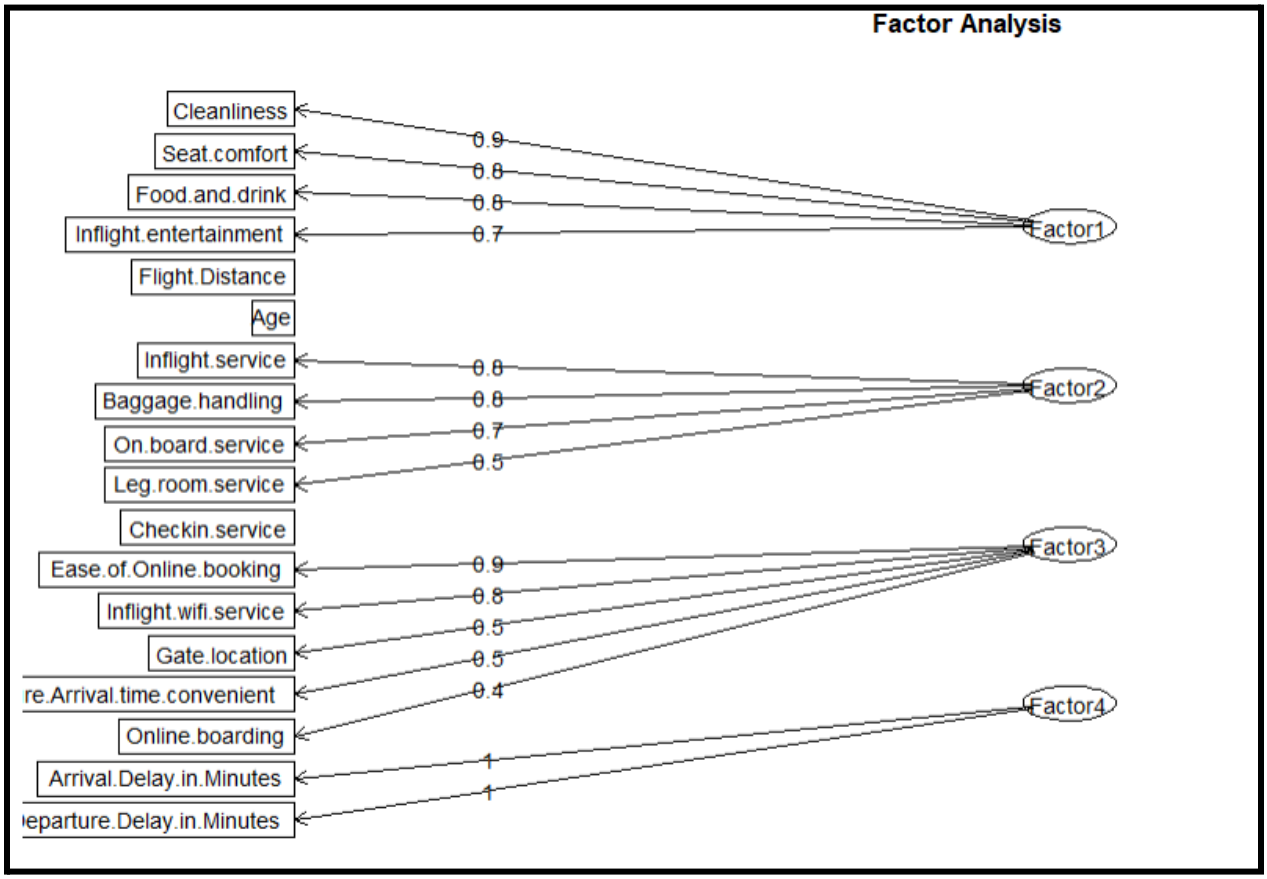
- **Variance explained:** 15.0%, 13.0%, 11.6% and 10.7%, for **50.4% cumulative**. That is moderate, and half the variance is unexplained.
- **Uniqueness:**
  - Low uniqueness means well explained: arrival delay (0.005), departure delay (0.063), ease of booking (0.115).
  - High uniqueness means poorly explained: Age (0.986), Flight Distance (0.973), Check-in (0.916). These load on no factor, so they are mostly independent of the service dimensions.
- **Varimax** gives uncorrelated factors with a simple structure, which makes them easy to interpret.
- **Goodness-of-fit test:** $\chi^2$ = 64,966.8 on 87 df, p = 0, so 4 factors are statistically “not sufficient”. With such a large n this test almost always rejects, but it does hint that more factors fit better.

```
fa<-factanal(x=scaled_data,factors = 5,scores = "regression")
scores =fa$scores
head(scores)
dim(scores)
```

```
> head(scores)
      Factor1    Factor2     Factor3    Factor4     Factor5
1  1.33848257  0.6271112  0.07814911  0.1153127 -0.46212494
2 -1.93246339 -0.4127490  0.19222027 -0.2844578  0.05611359
3  1.20918353  0.2994257 -0.81600703 -0.3528474  1.18656222
4 -0.96655172 -0.3322404  1.45136784 -0.1881138 -1.16001683
5  0.03104026 -0.3373321 -0.02505720 -0.3788285  1.32849759
6 -1.98742437  0.2784480 -0.35069691 -0.4261559 -0.48314993
> dim(scores)
[1] 103594      5
```

**#factor diagram**

fa.diagram(fa_model$loadings, cex = 0.9)




## Interpretation: Factor scores and diagram

- Refitting with 5 factors and `scores = "regression"` gives a 103,594 × 5 score matrix, one standardized score per passenger per factor.
- This turns 18 correlated variables into 5 compact predictors for the regression, which reduces multicollinearity.
- The factor diagram shows the 4-factor loadings graphically. It confirms the four groups above and that Age and Flight Distance are left out.

# Logistic regression

dataframe=cbind(data,scores)
dim(dataframe)

```
> dim(dataframe)
[1] 103594     30
```

```
model <- glm(
  satisfaction ~ Factor1 + Factor2 + Factor3 + Factor4  + Factor5 +
    Gender + Customer.Type + Age + Type.of.Travel + Class +
    Departure.Delay.in.Minutes + Arrival.Delay.in.Minutes,
  family="binomial", data=dataframe
)
```

summary(model)

```
Call:
glm(formula = satisfaction ~ Factor1 + Factor2 + Factor3 + Factor4 +
    Factor5 + Gender + Customer.Type + Age + Type.of.Travel +
    Class + Departure.Delay.in.Minutes + Arrival.Delay.in.Minutes,
    family = "binomial", data = dataframe)

Coefficients:
                                Estimate Std. Error z value Pr(>|z|)
(Intercept)                    8.968e+01  8.746e+00  10.253  < 2e-16 ***
Factor1                       -3.016e+00  3.552e-01  -8.492  < 2e-16 ***
Factor2                       -3.010e+00  3.996e-01  -7.534 4.91e-14 ***
Factor3                       -2.839e-01  5.785e-02  -4.908 9.19e-07 ***
Factor4                        2.297e+02  2.232e+01  10.294  < 2e-16 ***
Factor5                       -2.060e+00  3.120e-01  -6.603 4.02e-11 ***
GenderMale                     7.248e-02  1.893e-02   3.829 0.000129 ***
Customer.TypeLoyal Customer    1.759e+00  2.672e-02  65.842  < 2e-16 ***
Age                           -8.920e-03  6.909e-04 -12.909  < 2e-16 ***
Type.of.TravelPersonal Travel -2.685e+00  2.864e-02 -93.754  < 2e-16 ***
ClassEco                      -6.781e-01  2.275e-02 -29.803  < 2e-16 ***
ClassEco Plus                 -7.782e-01  3.815e-02 -20.396  < 2e-16 ***
Departure.Delay.in.Minutes    -5.095e-01  4.991e-02 -10.209  < 2e-16 ***
Arrival.Delay.in.Minutes      -5.441e+00  5.278e-01 -10.310  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 141768  on 103593  degrees of freedom
Residual deviance:  72454  on 103580  degrees of freedom
AIC: 72482

Number of Fisher Scoring iterations: 5
```

## Interpretation: Logistic regression (coefficient interpretation)

Odds ratio = e^coefficient. All predictors are significant (p < 0.001).

| Predictor | Estimate | Odds ratio | Meaning |
|---|---|---|---|
| Customer type: Loyal | +1.759 | ≈ 5.8 | Loyal customers have about 5.8× higher odds of being satisfied |
| Travel: Personal | −2.685 | ≈ 0.068 | About 93% lower odds than business travellers |
| Class: Eco | −0.678 | ≈ 0.51 | About half the odds of Business class |
| Class: Eco Plus | −0.778 | ≈ 0.46 | Even lower than Eco |
| Age | −0.0089 | ≈ 0.991 per year | About 8.5% lower odds per 10 years |
| Gender: Male | +0.072 | ≈ 1.08 | Statistically significant but a small effect |

**Model fit:**

- Null deviance is 141,768 and residual deviance is 72,454. The drop of about 69,300 for 13 predictors shows a big improvement over the null model.
- AIC is 72,482 and the model converged in 5 Fisher iterations.

**Strongest drivers:** travel type, customer type and class matter more than demographics like age and gender.

**# Accuracy**

```
prob <- predict(model, type = "response")
p <- ifelse(prob > 0.5, 1, 0)
m <- table(data$satisfaction, p)
Accuracy <- sum(diag(m)) / sum(m)
```

```
> Accuracy
[1] 0.8646254
```

```
pred<-ifelse(prob>0.5,"satisfied","netural or dissatisfed")
pred<-as.factor(pred)

#confusion matrix
conf_matrix<-table(Actual=data$satisfaction,Predicted=pred)
conf_matrix
```

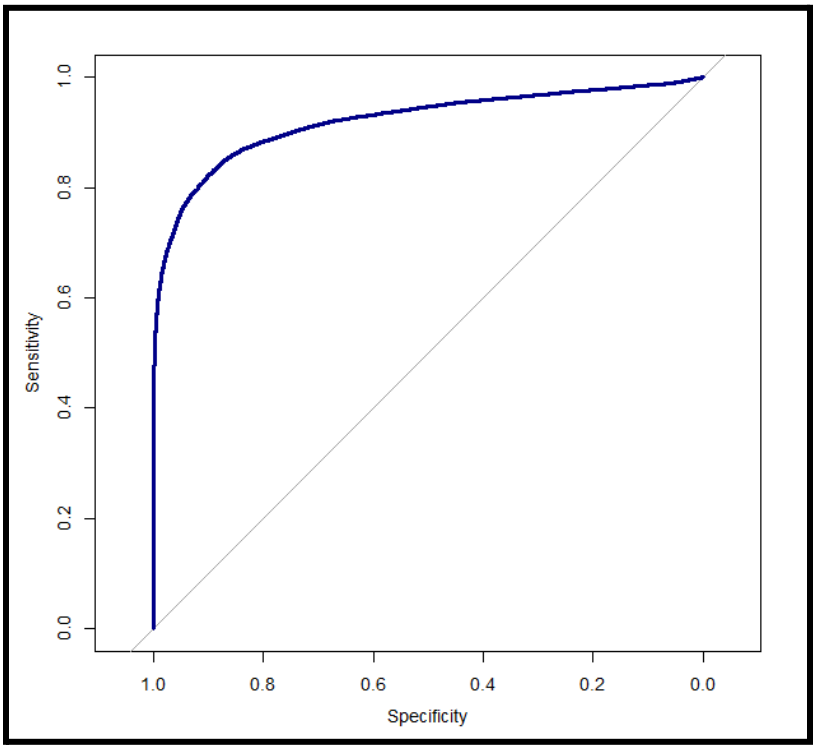
```
> conf_matrix
                        Predicted
Actual                   netural or dissatisfed satisfied
  neutral or dissatisfied                  52575      6122
  satisfied                                 7902     36995
```

```
corrplot(as.matrix(conf_matrix), is.corr = FALSE,  method = "color",addCoef.col = "black",
         tl.col = "black",col = colorRampPalette(c("white", "pink"))(200))
```




**#ROC plot**

```
roc_obj <- roc(data$satisfaction, prob)
plot(roc_obj, col="blue4", lwd=3)
```



auc(roc_obj)

```
> auc(roc_obj)
Area under the curve: 0.9201
```

## Interpretation: Accuracy, confusion matrix, ROC

**Confusion matrix:**

| | Pred. neutral/dissat. | Pred. satisfied |
|---|---|---|
| **Actual neutral/dissat.** | 52,575 (TN) | 6,122 (FP) |
| **Actual satisfied** | 7,902 (FN) | 36,995 (TP) |

- **Accuracy:** 0.8646, meaning 86.5% correct. The majority-class baseline is about 56.7%, so the gain is large.
- **Sensitivity (recall):** 36,995 / 44,897 ≈ **0.824**
- **Specificity:** 52,575 / 58,697 ≈ **0.896**
- **Precision:** 36,995 / 43,117 ≈ **0.858**
- **F1: ≈ 0.841**
- **AUC = 0.9201.** This is excellent discrimination: a randomly chosen satisfied passenger gets a higher predicted probability than a randomly chosen unsatisfied one 92% of the time. The ROC curve rises steeply toward the top-left and stays far above the diagonal.
- The model is slightly better at identifying dissatisfied passengers (specificity) than satisfied ones (sensitivity).